# E-Commerce Conversion Funnel and Marketing Campaign Analysis (SQL & BigQuery)

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**Tools:** PostgreSQL, BigQuery

## Introduction

This study uses the "bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*" dataset, which is an obfuscated version of the real Google Analytics 4 (GA4) data collected by Google from the Google Merchandise Store (the official store where Google branded products are sold).

This dataset contains the columns event_date, event_timestamp, event_name, event_parameters, user_pseudo_id (a temporary ID defined separately for each user session), user_id (a permanent ID identifying the unique user), and device_category, and each row represents a user action.

**1. Exercise:**

**1.1 SQL Query:**

```
SELECT
timestamp_micros(event_timestamp) as event_timestamp,
user_pseudo_id,
(SELECT value.int_value FROM UNNEST(event_params) WHERE key = 'ga_session_id') as session_id,
event_name,
geo.country as country,
device.category as device_category,
traffic_source.source as source,
traffic_source.medium as medium,
traffic_source.name as campaign
FROM
`bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`
WHERE
_table_suffix between '20210101' and '20211231'
and event_name in
('session_start', 'view_item', 'add_to_cart', 'begin_checkout', 'add_shipping_info', 'add_payment_info', 'purchase')
ORDER BY
event_timestamp, user_pseudo_id, session_id;
```

**1.2 Results Obtained:** User actions have been simplified and compiled by specifying event_timestamp, user_pseudo_id, session_id, event_name, country, device_category, source, medium, and campaign attributes on a user-by-user basis.

To represent the "Purchase Funnel" structure in its purest form on an e-commerce site, the actions have been limited to 'session_start', 'view_item', 'add_to_cart', 'begin_checkout', 'add_shipping_info', 'add_payment_info', and 'purchase'.

| | A | B | C | D | E | F | G | H | I |
|---|---|---|---|---|---|---|---|---|---|
| 1 | event_timestamp | user_pseudo_id | session_id | event_name | country | device_category | source | medium | campaign |
| 2 | 2021-01-01 00:00:08 | 63015181.07 | 8529617896 | session_start | India | mobile | google | organic | (organic) |
| 3 | 2021-01-01 00:00:43 | 2688110.978 | 5376800584 | view_item | United States | desktop | google | organic | (organic) |
| 4 | 2021-01-01 00:00:44 | 13725727.95 | 1274854359 | session_start | United States | desktop | google | cpc | <Other> |
| 5 | 2021-01-01 00:01:16 | 1112823.848 | 4251926878 | view_item | India | desktop | (direct) | (none) | (direct) |
| 6 | 2021-01-01 00:01:31 | 2688110.978 | 5376800584 | view_item | United States | desktop | google | organic | (organic) |
| 7 | 2021-01-01 00:01:56 | 2688110.978 | 5376800584 | view_item | United States | desktop | google | organic | (organic) |
| 8 | 2021-01-01 00:02:31 | 2688110.978 | 5376800584 | view_item | United States | desktop | google | organic | (organic) |
| 9 | 2021-01-01 00:02:51 | 16012688.82 | 4234093164 | session_start | China | mobile | shop.googlemerchar | referral | (referral) |
| 10 | 2021-01-01 00:03:02 | 6272186.264 | 8867381478 | session_start | China | desktop | google | cpc | <Other> |
| 11 | 2021-01-01 00:03:07 | 6272186.264 | 8867381478 | view_item | China | desktop | google | cpc | <Other> |
| 12 | 2021-01-01 00:04:36 | 7949252.633 | 7586993078 | session_start | Canada | mobile | (direct) | (none) | (direct) |
| 13 | 2021-01-01 00:04:39 | 5191443912 | 7452070873 | session_start | United States | mobile | google | organic | (organic) |
| 14 | 2021-01-01 00:05:58 | 6605976219 | 1348358504 | session_start | France | desktop | google | organic | (organic) |
| 15 | 2021-01-01 00:06:34 | 6605976219 | 1348358504 | view_item | France | desktop | google | organic | (organic) |
| 16 | 2021-01-01 00:06:49 | 6661636.522 | 1565208642 | session_start | United States | desktop | (direct) | (none) | (direct) |
| 17 | 2021-01-01 00:07:07 | 1112823.848 | 4251926878 | begin_checkout | India | desktop | (direct) | (none) | (direct) |
| 18 | 2021-01-01 00:07:07 | 1112823.848 | 4251926878 | add_shipping_info | India | desktop | (direct) | (none) | (direct) |
| 19 | 2021-01-01 00:07:07 | 1112823.848 | 4251926878 | begin_checkout | India | desktop | (direct) | (none) | (direct) |
| 20 | 2021-01-01 00:07:08 | 1112823.848 | 4251926878 | begin_checkout | India | desktop | (direct) | (none) | (direct) |
| 21 | 2021-01-01 00:07:22 | 9115846.176 | 935901267 | session_start | United States | mobile | (direct) | (none) | (direct) |
| 22 | 2021-01-01 00:07:31 | 43729465.24 | 5236946313 | session_start | South Korea | mobile | <Other> | <Other> | <Other> |
| 23 | 2021-01-01 00:07:43 | 1112823.848 | 4251926878 | view_item | India | desktop | (direct) | (none) | (direct) |
| 24 | 2021-01-01 00:07:55 | 1112823.848 | 4251926878 | begin_checkout | India | desktop | (direct) | (none) | (direct) |
| 25 | 2021-01-01 00:07:55 | 1112823.848 | 4251926878 | add_shipping_info | India | desktop | (direct) | (none) | (direct) |
| 26 | 2021-01-01 00:07:55 | 1112823.848 | 4251926878 | begin_checkout | India | desktop | (direct) | (none) | (direct) |
| 27 | 2021-01-01 00:07:56 | 1112823.848 | 4251926878 | begin_checkout | India | desktop | (direct) | (none) | (direct) |

**2.Exercise:**

## 2.1 SQL Query:

*- Only the rows containing the session_start action and their event_date, user_pseudo_id, source, medium, and campaign columns were extracted to create a CTE named session_starts.*

```
WITH session_starts as

(

SELECT

date(timestamp_micros(event_timestamp)) as event_date,

user_pseudo_id,

(SELECT value.int_value FROM unnest(event_params) WHERE key='ga_session_id') as
session_id,

traffic_source.source as source_,

traffic_source.medium as medium,

traffic_source.name as campaign

FROM

`bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`

WHERE

_table_suffix between '20210101' and '20211231'

AND event_name = 'session_start'

),
```

*- The CTE named specific_events was created by extracting the properties and actions of add_to_cart, begin_checkout, and purchase, which are other actions whose conversions will be analyzed.*

```
specific_events as

(

SELECT

user_pseudo_id,

(SELECT value.int_value FROM unnest(event_params) WHERE key = 'ga_session_id') as
session_id,

event_name
```

```
FROM
`bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`
WHERE
_table_suffix between '20210101' and '20211231'
AND event_name in ('add_to_cart', 'begin_checkout', 'purchase')
),
```

*Then, the previously created CTEs were joined using the join command, and a CTE named `useful_table` was created, containing temporary columns that take values of 1 and 0 depending on whether the relevant actions are present in that row or not, to be summed using the sum command.*

```
useful_table as
(
SELECT
s.event_date,
s.source_,
s.medium,
s.campaign,
s.user_pseudo_id,
s.session_id,
```

*--This trio gives us a 1 or 0 depending on whether each row we get from the s table has the corresponding event_name.*

```
MAX(case when e.event_name = 'add_to_cart' then 1 else 0 end) as cart,
MAX(case when e.event_name = 'begin_checkout' then 1 else 0 end) as checkout,
MAX(case when e.event_name = 'purchase' then 1 else 0 end) as purchase
FROM
session_starts s
LEFT JOIN
specific_events e
ON s.user_pseudo_id = e.user_pseudo_id and s.session_id = e.session_id
GROUP BY
```

```
s.event_date, s.source_, s.medium, s.campaign, s.user_pseudo_id, s.session_id

)

--In the final query where we obtain the results table, we perform the transformations using the 0-1 columns that indicate the presence or absence of the relevant action, which we created in the CTE named "useful table". By combining the User_id and Session_id actions as strings, we obtain a unique identifier for each session for each user.

SELECT

event_date,

source_,

medium,

campaign,

-- User_ID and session_ID are combined to identify unique sessions.

COUNT(DISTINCT CONCAT(CAST(user_pseudo_id AS STRING), ':', CAST(session_id AS STRING))) as user_sessions_count,

--The sum of the 1s in the columns we created earlier gives the number of the relevant event.

ROUND(SAFE_DIVIDE(SUM(cart), COUNT(DISTINCT CONCAT(CAST(user_pseudo_id AS STRING), ':', CAST(session_id AS STRING)))),4) as visit_to_cart,


ROUND(SAFE_DIVIDE(SUM(checkout), COUNT(DISTINCT CONCAT(CAST(user_pseudo_id AS STRING), ':', CAST(session_id AS STRING)))),4) as visit_to_checkout,


ROUND(SAFE_DIVIDE(SUM(purchase), COUNT(DISTINCT CONCAT(CAST(user_pseudo_id AS STRING), ':', CAST(session_id AS STRING)))),4) as visit_to_purchase,

FROM

useful_table

GROUP BY

event_date, source_, medium, campaign

ORDER BY

event_date, source_, medium, campaign, user_sessions_count DESC;
```

**2.2 Onbtained Result:** Based on the channel used to access the site, the date, and the number of unique sessions opened on that channel and date, we can see conversion rates from visits to add to cart, checkout, and purchase. By comparing these, strategic decisions can be made.

| | A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|---|
| 1 | event_date | source_ | medium | campaign | user_sessions_count | visit_to_cart | visit_to_checkout | visit_to_purchase |
| 2 | 2021-01-01 | (data deleted) | (data deleted) | (data deleted) | 85 | 0.0588 | 0.0235 | 0.0235 |
| 3 | 2021-01-01 | (data deleted) | (data deleted) | <Other> | 3 | 0 | 0 | 0 |
| 4 | 2021-01-01 | (direct) | (none) | (direct) | 564 | 0.0408 | 0.0142 | 0.0089 |
| 5 | 2021-01-01 | <Other> | (data deleted) | (data deleted) | 2 | 0 | 0 | 0 |
| 6 | 2021-01-01 | <Other> | <Other> | <Other> | 377 | 0.0186 | 0.0159 | 0 |
| 7 | 2021-01-01 | <Other> | organic | (organic) | 64 | 0.0156 | 0 | 0 |
| 8 | 2021-01-01 | <Other> | referral | (referral) | 211 | 0.0284 | 0.0047 | 0.0047 |
| 9 | 2021-01-01 | google | cpc | <Other> | 103 | 0.0194 | 0.0097 | 0.0097 |
| 10 | 2021-01-01 | google | organic | (organic) | 746 | 0.0268 | 0.0107 | 0.004 |
| 11 | 2021-01-01 | shop.googlemerchar | referral | (referral) | 134 | 0.0224 | 0 | 0 |
| 12 | 2021-01-02 | (data deleted) | (data deleted) | (data deleted) | 96 | 0.0417 | 0.0521 | 0.0104 |
| 13 | 2021-01-02 | (data deleted) | (data deleted) | <Other> | 3 | 0 | 0 | 0 |
| 14 | 2021-01-02 | (direct) | (none) | (direct) | 633 | 0.0221 | 0.0079 | 0.0032 |
| 15 | 2021-01-02 | <Other> | <Other> | <Other> | 428 | 0.0397 | 0.014 | 0.0047 |
| 16 | 2021-01-02 | <Other> | organic | (organic) | 72 | 0 | 0 | 0 |
| 17 | 2021-01-02 | <Other> | referral | (referral) | 281 | 0.0214 | 0.0107 | 0.0071 |
| 18 | 2021-01-02 | google | cpc | <Other> | 147 | 0.0068 | 0 | 0 |
| 19 | 2021-01-02 | google | organic | (organic) | 896 | 0.0212 | 0.0134 | 0.0067 |
| 20 | 2021-01-02 | shop.googlemerchar | referral | (referral) | 149 | 0.0201 | 0.0067 | 0.0067 |
| 21 | 2021-01-03 | (data deleted) | (data deleted) | (data deleted) | 92 | 0.0761 | 0.0435 | 0.0326 |
| 22 | 2021-01-03 | (data deleted) | (data deleted) | <Other> | 4 | 0 | 0 | 0 |
| 23 | 2021-01-03 | (direct) | (none) | (direct) | 634 | 0.0174 | 0.0079 | 0.0032 |
| 24 | 2021-01-03 | <Other> | (data deleted) | (data deleted) | 2 | 0 | 0 | 0 |
| 25 | 2021-01-03 | <Other> | <Other> | <Other> | 476 | 0.0273 | 0.0168 | 0.0084 |
| 26 | 2021-01-03 | <Other> | organic | (organic) | 91 | 0.011 | 0 | 0 |
| 27 | 2021-01-03 | <Other> | referral | (referral) | 280 | 0.0107 | 0.0036 | 0.0036 |

## 3.Exercise:

## 3.1 SQL Query:

*By extracting the page_path, session-id, and user_pseudo_id from only the rows containing the session_start action, information about the users who started the session can be obtained.*

```
WITH session_start_info AS

(

SELECT

(SELECT value.int_value FROM UNNEST(event_params) WHERE key = 'ga_session_id') as session_id,

user_pseudo_id,

regexp_extract((SELECT value.string_value FROM UNNEST(event_params) WHERE key = 'page_location'),r'https?://[^/]+(.*)')

as page_path

FROM

`bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`WHERE

_table_suffix between '20200101' and '20201231'

AND event_name = 'session_start'

),

--Information about the purchased sessions is obtained.

puchase_sessions AS

(

SELECT

(SELECT value.int_value FROM UNNEST(event_params) WHERE key = 'ga_session_id') as session_id,

user_pseudo_id

FROM

`bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*`

WHERE

_table_suffix between '20200101' and '20201231'
```

```
AND event_name = 'purchase'
)
--The two CTEs created for the results table are combined.
SELECT
s.page_path,
count(distinct s.session_id) as unique_session_count,
count(p.session_id) as purchase_count,
round((safe_divide(count(distinct p.session_id),count(distinct s.session_id))),2) as
session_puchase_count
FROM
session_start_info s
LEFT JOIN
puchasing_sessions p
ON s.session_id = p.session_id
AND s.user_pseudo_id = p.user_pseudo_id
WHERE
s.page_path is not null
AND s.page_path != ''
GROUP BY
s.page_path
ORDER BY
uniqu_session_count DESC;
```

**3.2.Obtained Result:** The number of sessions visited and opened from a given page path, the number of sessions resulting in a purchase, and the conversion rate from total sessions to purchases are obtained. This allows for analysis of which page paths have a higher purchase success rate.

| | A | B | C | D |
|---|---|---|---|---|
| 1 | page_path | unique_session_cou | purchase_count | session_purchase_cc |
| 2 | / | 107491 | 2421 | 0.02 |
| 3 | /Google+Redesign/A | 22248 | 132 | 0.01 |
| 4 | /Google+Redesign/S | 16933 | 61 | 0 |
| 5 | /Google+Redesign/A | 11321 | 1 | 0 |
| 6 | /store.html | 9168 | 93 | 0.01 |
| 7 | /Google+Redesign/A | 4751 | 43 | 0.01 |
| 8 | /Google+Redesign/L | 3733 | 33 | 0.01 |
| 9 | /Google+Redesign/A | 3662 | 147 | 0.03 |
| 10 | /signin.html | 3166 | 73 | 0.02 |
| 11 | /Google+Redesign/S | 2799 | 152 | 0.05 |
| 12 | /basket.html | 2648 | 251 | 0.07 |
| 13 | /Google+Redesign/L | 2631 | 19 | 0 |
| 14 | /store-policies/freque | 2312 | 4 | 0 |
| 15 | /Google+Redesign/C | 2297 | 63 | 0.02 |
| 16 | /Google+Redesign/N | 2001 | 48 | 0.02 |
| 17 | /Google+Redesign/A | 1711 | 5 | 0 |
| 18 | /Google+Redesign/S | 1549 | 8 | 0.01 |